

## Measurement Validity

- Measurement error → bad data → worthless results
- Is the IV properly manipulated?
- Is the DV properly measured?
- Are the values we have for **every** measure/behavior correct?
- Observational, Self-report or Trace?
- Primary or Archival data?

**Every Characteristic/Behavior/Procedure is either...**

|             | Constant | Variable |
|-------------|----------|----------|
| Measured    | 1        | 2        |
| Manipulated | 3        | 4        |

## IV → DV causes

- **Temporal Precedence**
- **Reliable statistical relationship**
- **No alternative hypotheses/confounds**

measurement classification → study role and equivalence

## Statistical Conclusion Validity

- IV & DV can't be causally related if not statistically related
- Statistical significance tests
- Programmatic Research – novel RH tests, replication & convergence

## Characteristic/Behavior/Procedure Plays a "Role" in a Study

| Role                        | Equivalence                    |
|-----------------------------|--------------------------------|
| Causal Variable (IV)        | 4 – Ongoing Eq                 |
| Effect Variable (DV)        | 2 – Ongoing Eq                 |
| <b>Control Constant</b>     | 1 – Initial Eq; 3 – Ongoing Eq |
| <b>Control Variable</b>     | 2 – Initial Eq; 4 – Ongoing Eq |
| <b>Confounding Variable</b> | 2 – Initial Eq; 4 – Ongoing Eq |

## External Validity

### Population – Participant Sampling

|                   |                                      |
|-------------------|--------------------------------------|
| Stage             | Choice or outcome                    |
| Target population | Complete or Purposive                |
| Sampling Frame    | Researcher-selected or Self-selected |
| Selected Sample   | Simple or Stratified                 |
| Data Sample       | Attrition                            |

### Setting

- Laboratory, Structured or Field?

### Task/Stimulus

- Familiar/Representative or Unfamiliar/Control?

### Societal/Temporal

- Relationships among variables change over time in a society

## Choices we make influence Internal and External Validity !!

**Participants**  
representation vs. control

←-----→

**Setting**  
representation vs. control

←-----→

**Task-Stimulus**  
representation vs. control

←-----→

## Internal Validity Design

| Design          | BG | WG |
|-----------------|----|----|
| True Experiment | ✓  | ✓  |
| Non-experiment  | X  | X  |

**Legend:** ✓ = design supports causal interpretation; X = design does not support causal interpretation.

### Initial Equivalence – Participant Assignment

- RA of individual participants by the researcher before manipulation of the IV – best but not a guarantee
- Without proper RA all subject variables are potential confounds
- Subject constants can't be confounding variables
- Subject variables that are equivalent across IV conditions are control variables
- Subject variables that are nonequivalent across IV conditions are confounding variables – even if RA was used (remember RA doesn't always work)

### Ongoing Equivalence – Procedural Standardization

- Only the IV is different across IV conditions
- Procedural constants cannot be confounding variables
- Procedural variables that are equivalent across IV conditions are control variables
- Procedural variables that are nonequivalent across IV conditions are confounding variables
- Ongoing equivalence is harder to maintain in field settings
- Ongoing equivalence is harder to maintain during longer procedures